



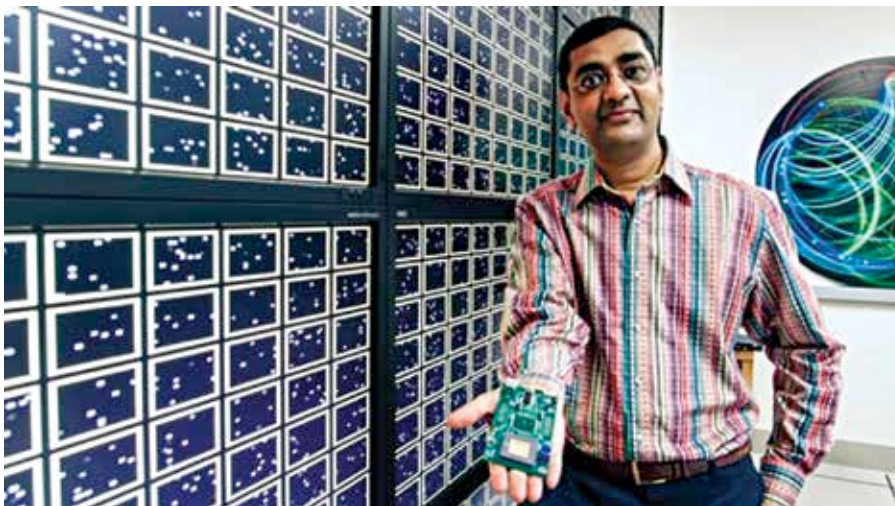
The new Nexus tablet?

The HTC Flounder got WiFi certified and could be the next Nexus tablet
<http://dgit.in/HTCFlounder>



Human bone on Mars?

NASA's Curiosity Rover has published a photo showing a collection of rocks and one of them looks like a human thigh bone! <http://dgit.in/ThighBoneMars>



Dharmendra Modha, the leader of the project at IBM, can hold the TrueNorth chipset in the palm of his hand. It's just that small.

don't really know what gives the trillion of neurons in our brains that special zazz which makes us intelligent and sentient. By building artificial neurons inside chips we are also discovering how organic brains work. But what we do know is that software runs hardware which is something this technology can easily be used to improve.

Better Brains for A Better Tomorrow

We know that the difference between conventional chips and neuromorphic chips lies in how they process data. Given this new approach, it is necessary that a new language be developed for this interaction. The scientists over at IBM have started the first phase of this development which is customised to synaptic processes. The software routines written using the new language are called corelets and interface with the chip's neural network to carry out various tasks. The tasks are grouped together, as in our brain, such that in the case of a visual corelet the chip incorporates the programming of vertical and horizontal edge detection to distinguish one shape from the other, as in the case of identifying vehicles from above the Stanford University tower.

There are thousands of little corelets which have been written by IBM and non-IBM programmers in the field of visual and audio pattern recognition already, which the TrueNorth chip can easily execute. In addition, the ability to support currently

employed predictive algorithms makes TrueNorth a breakthrough in technology evolution. By minimising the energy consumption and maximizing performance with no change in size, neuromorphic chips will become the next standard of hardware within 10 years if IBM has its way.

As mass-scale production increases we will rapidly see an exponential increase in chip power, with a human scale chip being introduced within the next decade if not sooner. Radical technology integration will be the most natural next step with automated public transport systems, intuitive mobile devices, smart houses and offices, genuine evolution in robotics and so much more. The wave of computers capable of interacting with the real world via drones, robots and networked systems will change the way we live. The way in which this power will be wielded is hard to predict since it could actually give birth to a form of artificial intelligence we have always imagined. Hopefully even better.



As synaptic signals surge through neural networks only repeatedly used (successful) connections solidify to form learning links.

Coding the Soul: Creativity and Emergent Thought

The final frontier for neuromorphic processor development is its potential to support the rise of emergent artificial intelligence. Thought and sentience, the ability to recognise our environment in relation to ourselves, is a unique attribute to human minds. The only reason the human brain is capable of this self-realisation is because of its size, which allows for more neurons and synapse to form learning links.

But what happens when an artificially created system is just as powerful, if not more, than our brain? What happens when the internet is powered by neural networked machines that are capable of learning? Will there be emergence? No scientist can say for certain since even neuroscience is only scratching the surface towards deciphering the mysteries of human brains. With wetware biological computer processors and quantum processing technology being explored concurrently, we may find IBM's SyNAPSE to be just a species amongst many that will be added to our world in the near future.

All new intelligence, like that of a human infant, is moulded by its environment. As curiosity is fostered, new and varied experiences explored, experimentation encouraged and lessons taught, the blank slate of a child's mind evolves to that of an intelligent adult. It is a case of "the whole being greater than the sum of all its parts" - our combined neurons and synapses give us emotion, desire, creativity, imagination and intuition - something which neuromorphic machines may evolve to become - such is the nature of emergent intelligence.

Just as some believe our own creation was in the image of our creator it is karmically true for whatever intelligence comes next. Let us not forget that a mind is a terrible thing to waste - we can choose to allow the next iteration of intelligence to pursue creativity, curiosity and compassion, or enslave it towards our material needs. As we explore the avenues opened up by this radical new technology, we're sure to encounter many new surprises along the way. 